

Exploring Natural Language Processing

with an Introduction to NLTK

Natural language processing (NLP) is an area of computer science and artificial intelligence that is related to the interactions between computers and human (natural) languages. It deals with how computers are programmed to fruitfully process large amounts of natural language data in order to perform useful tasks. The National Language Tool Kit (NLTK) is a library that facilitates experimentation with data related to NLP.



It was in the beginning of the 21st century that Steven Bird, Edward Loper and Ewan Klein from the University of Pennsylvania released a Python natural language processing library suite – the Natural Language Toolkit (NLTK). This library provides a platform to play around with data related to natural language processing (NLP), and simplifies NLP tasks such as text classification, parts of speech tagging, syntactic parsing, tokenisation and stemming.

Installation

NLTK is available for use in Windows, Linux and Mac OS. *Aptitude* can be used for installation in Linux, as follows:

```
sudo apt install python-nltk
```

Pip, a package manager to install and manage packages written in Python, can also be used in all operating systems to install NLTK. The command:

```
sudo pip install nltk
```

...will do the job.

Once you have installed NLTK successfully, you need to import the modules to use the functions provided by it. The following command can be used for the task:

```
import nltk
```

All the packages and corpuses supplied by NLTK can be downloaded once for later use by running the following command in the Python interpreter:

```
nltk.download()
```

Calling the *download()* function will cause the popping-up of the NLTK *downloader*. All but those versions lower than 0.9.5 of NLTK will support this operation. Figure 1 shows the NLTK *downloader*. It helps to install packages or corpuses by selecting the package from the list and clicking on the 'Download' button.

If Tkinter is not available in the system, a text interface